

EXP 1.1

CODE:

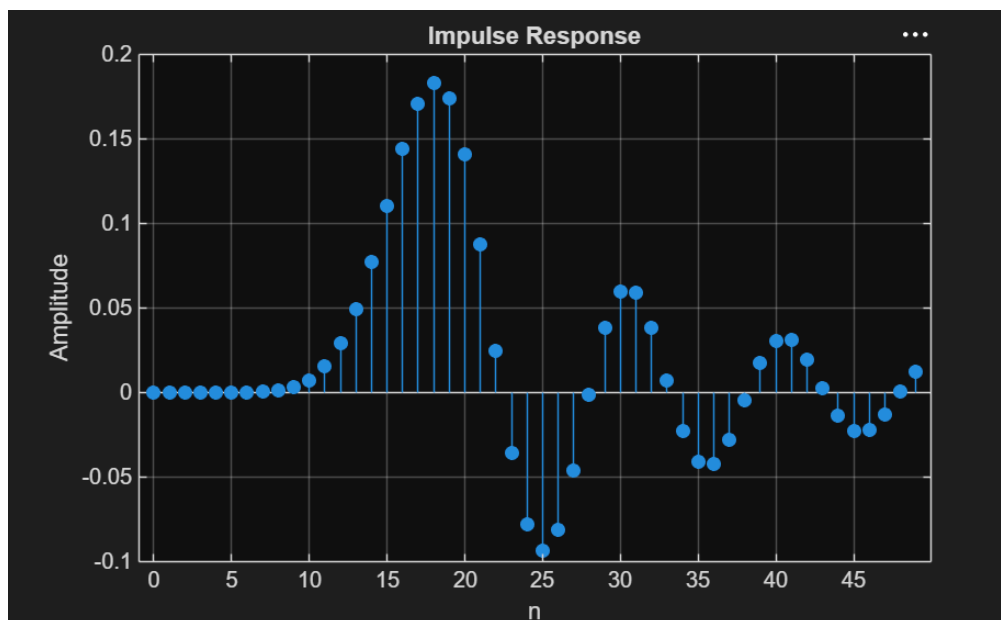
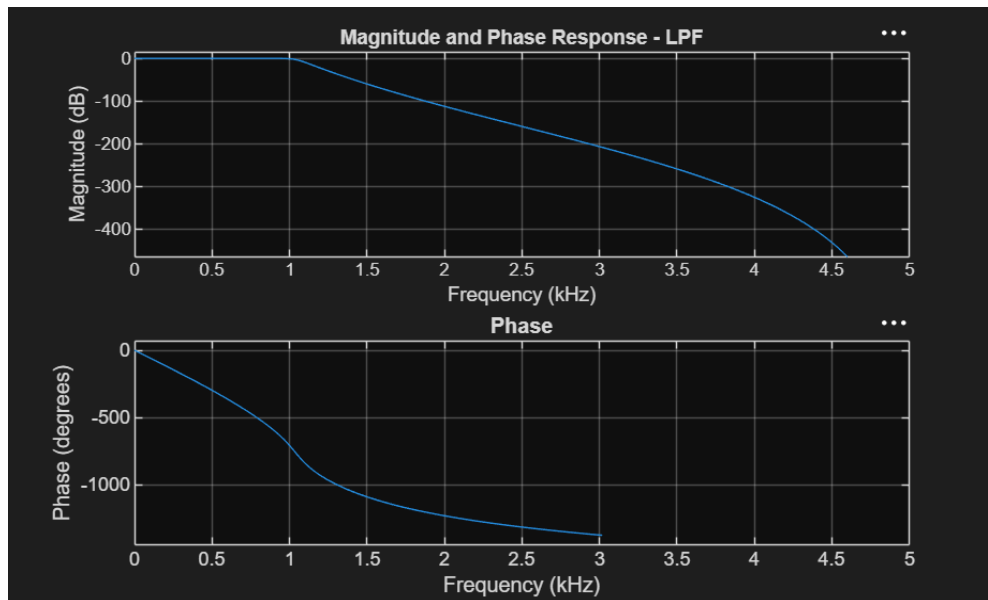
```
%% Exp 1.1: Design and Analysis of a Butterworth IIR Low-Pass Filter
clc; clear; close all;
%% Filter Specifications
Fs = 10000; % Sampling frequency (Hz)
Fp = 1000; % Passband edge frequency (Hz)
Fst = 1500; % Stopband edge frequency (Hz)
Rp = 1; % Passband ripple (dB)
Rs = 60; % Stopband attenuation (dB)
%% Normalize frequencies w.r.t. Nyquist frequency
Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);
%% Determine minimum order and cutoff frequency
[N,Wn] = buttord(Wp,Ws,Rp,Rs);
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
%% Design the Butterworth Low-Pass Filter
[b,a] = butter(N,Wn,'low');
%% Magnitude and Phase Response
figure;
freqz(b,a,1024,Fs);
title('Magnitude and Phase Response - LPF');
%% Impulse Response
figure;
[h,n] = impz(b,a,50);
stem(n,h,'filled'); grid on;
title('Impulse Response'); xlabel('n'); ylabel('Amplitude');
%% Step Response
figure;
stepz(b,a,50);
title('Step Response');
%% Group Delay
figure;
grpdelay(b,a,1024,Fs);
title('Group Delay');
%% Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot - LPF');
%% Test Signal: Low-frequency + High-frequency components
t = 0:1/Fs:0.05;
```

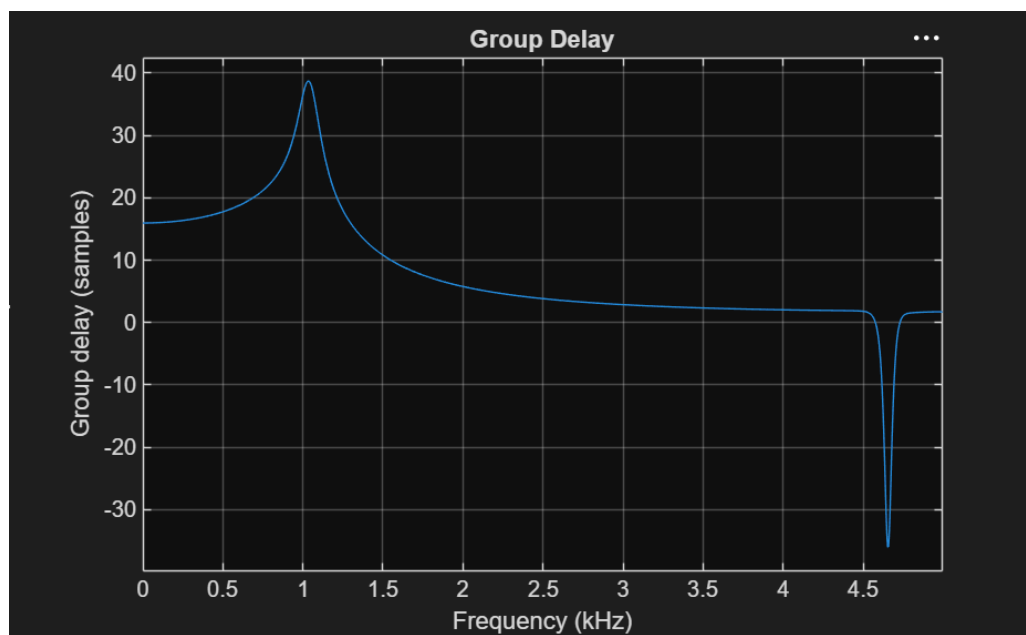
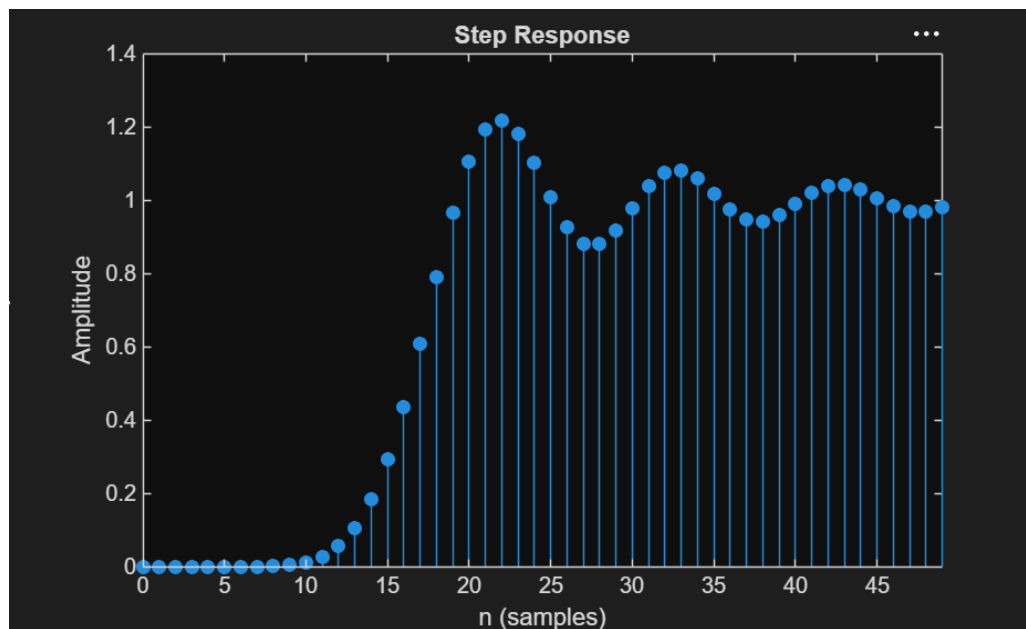
```

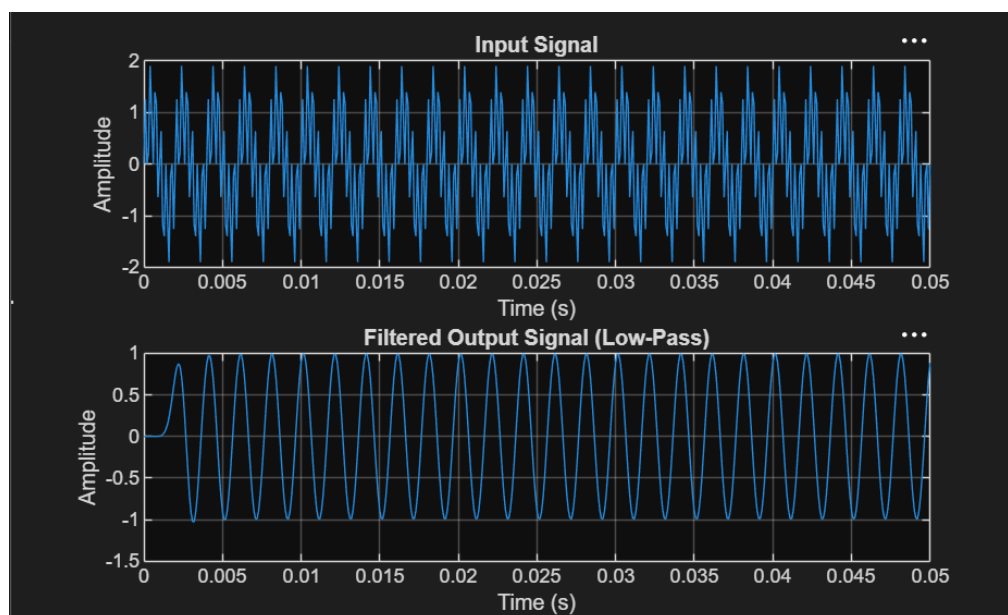
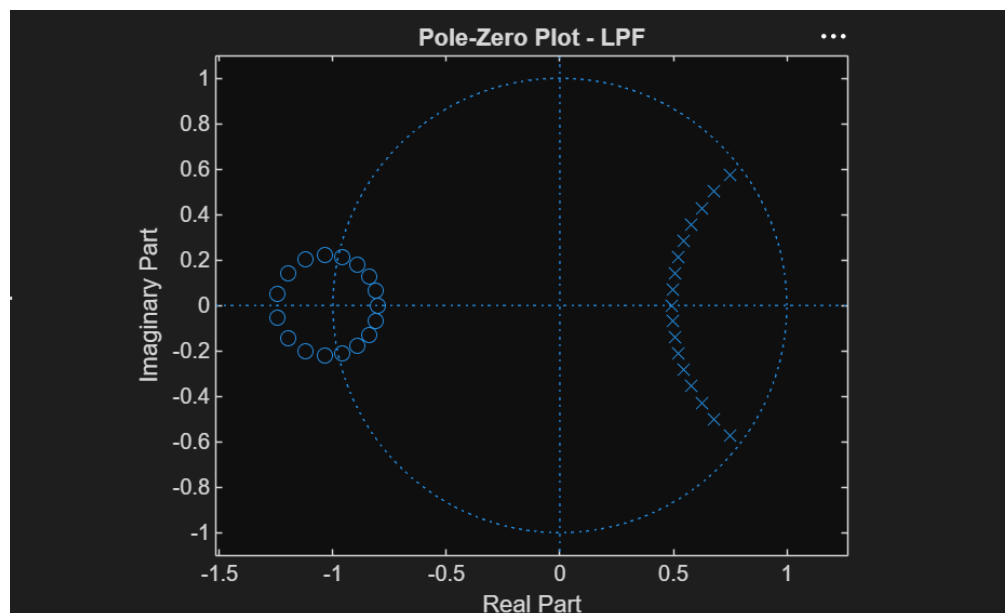
x = sin(2*pi*500*t) + sin(2*pi*3000*t); % 500 Hz (pass) + 3000 Hz (reject)
y = filter(b,a,x);
figure;
subplot(2,1,1); plot(t,x); grid on;
title('Input Signal'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,1,2); plot(t,y); grid on;
title('Filtered Output Signal (Low-Pass)'); xlabel('Time (s)'); ylabel('Amplitude');

```

OUTPUT:







EXP 1.2

CODE:

```
clc; clear;
Fs=10000; Fp=3000; Fst=2000;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn,'high');
% Magnitude & Phase
freqz(b,a)
% Impulse Response
figure
[h,n]=impz(b,a,50);
stem(n,h), grid on
title('Impulse Response')
% Step Response
figure
stepz(b,a)
title('Step Response')
% Group Delay
figure
grpdelay(b,a)
title('Group Delay')
% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

OUTPUT:

